

SUDOKU PUZZLE SOLVER PROGRAM IN JAVASCRIPT

■ BUKYA SRIDHAR

Sudoku is a logic-based number puzzle that does not need calculation but mental ability and analysis to solve it. Popularised by a Japanese puzzle company in 1986, it uses a grid that has to be filled in such a way that all the cells in it should have digits occurring only once—row-wise as well as column-wise.

Earlier, we have published 'Sudoku Generator and Solver in C++' article in the June 2007 issue. This program is a Sudoku solver coded in JavaScript, which displays the solution when you fill in numbers in the matrix cells and click on the Solve button. It is a handy tool to get quick solutions to various Sudoku matrices that are normally published in newspapers.

As JavaScript is easy to program and understand, it has been used for coding. Moreover, unlike Java, it does not need any special application or IDE (integrated development environment) for processing and compilation. It requires only Internet Explorer, which is pre-installed in almost every Windows-based computer.

The program has been so designed that it can solve simple problems to complex ones. First, a simple Sudoku solver with 3×3 matrix (as shown in Fig. 1) is designed. This is followed by Sudoku solvers with 6×6 matrix (as shown in Fig. 2) and 9×9 matrix (Fig. 3), respectively. Just by changing the cells, and a little bit of calculation, one can extend this to nth number of matrices. As the logic does not work for alphabets, it has been restricted to numbers only and permutations work on symmetric matrix.

The purpose of describing various matrices is only to show the flexibility

of the programming in JavaScript (JS). The only problem in this programming is that one needs to have a clear transparency in coding. Missing even a single cell results in error, hence the division and modulo (remainder) has to be set such that there exists no bug in the program.

Considering the 9×9 grid matrix, let's see how the program works.

1. The code checks for the valid entry, i.e., it accepts only numbers from 1 through 9 (0 or alphabets are not accepted). This function is taken care of

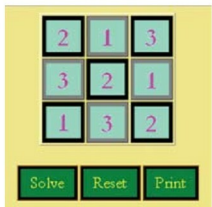


Fig. 1: 3×3 matrix

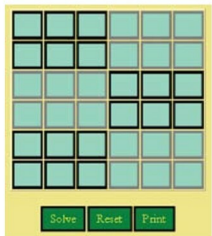


Fig. 2: 6×6 matrix



by the `checkIt` (event) function, which takes the input parameters. (Here 'event' is the one sent by `onkeypress` function.)

2. The `onkeypress` function reads the key code when you press a key from the keyboard.

3. When you click a grid for input, the program checks the key code and matches it with the one given in the code. If the input range is equal to the coded range, it displays the number in the grid.

4. The rows and columns of the grid are divided by using the `Math.floor` function. Every Sudoku has 3×3 equal sub-grids, as indicated by black border lines in all the figures shown here. Hence, a 9×9 matrix has a total of 81 grids.

Some important functions of the code are as under:

`solveIt()` function solves a puzzle according to the input code from the user.

`showarray()` function displays the

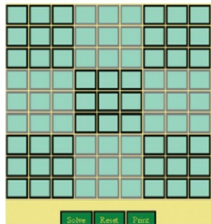


Fig. 3: 9×9 matrix